Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Backend engineer with 6+ years building scalable APIs, data pipelines, and search systems in production real estate software. Built 20+ features at Entrata -- a real estate SaaS platform with 100K+ monthly active users, the same domain as Realtor.com's search -- including async processing pipelines, SQL-optimized data layers, and AWS-based infrastructure. Built search and retrieval systems in personal projects: Leasel (hybrid TF-IDF + semantic search pipeline over real estate documents) and AgentProbe (ChromaDB vector search backend, 15-endpoint FastAPI). Currently shipping Node.js/Python services and AWS data pipelines at Intuit. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Backend APIs & Microservices | Search Pipelines | Elasticsearch | Node.js & Python | TypeScript & React | AWS (S3, EC2, Lambda) | Docker & Kubernetes | CI/CD & DevOps | SQL Query Optimization | Real Estate Domain | Agile/Scrum (120+ sprints) | Production Scale

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · Virginia, USA

- Develop **backend APIs and data pipelines** (Python/FastAPI) for financial data summaries and natural-language queries, with React/TypeScript front ends deployed to AWS (S3, EC2, Lambda, API Gateway) via GitHub Actions CI/CD.
- Integrate Anthropic Claude into Python services for intelligent query answering; ground LLM responses in customer financial data via retrieval pipelines so answers trace back to source records.
- Build ML pipelines (scikit-learn, XGBoost) from data ingestion through feature engineering and prediction APIs; add evaluation and regression tests to catch quality regressions before release.
- Containerize services with Docker and ship through GitHub Actions CI/CD to AWS with logging and monitoring for production visibility.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Real estate SaaS, 100K+ monthly active users

- Delivered **20+ backend features end-to-end** (Python, Node.js, Java, TypeScript, React, PostgreSQL) across 120+ Agile sprints in a multi-tenant real estate platform -- same domain as Realtor.com Search.
- Built a **RabbitMQ-based async processing system** for high-volume rent updates across thousands of tenant records, with retry logic and idempotency guarantees for reliable at-scale delivery.
- Optimized **200+ SQL queries** using execution plans, indexing, and partitioning -- 30% performance gain across complex real estate data schemas (eligibility, financials, document workflows).
- Maintained the HUD income-limit data pipeline for 7 years -- ingesting federal data into ~180,000 domain records annually with full validation, reconciliation, and rollback; zero data incidents.
- AWS S3 migration with Libsodium encryption for 100% PHI compliance; Docker microservices redesign (50% better scalability, 30% faster deployments); reviewed 300+ pull requests.
- Built Python/Bash automation eliminating 80% of manual operational workload; internal debugging tools reducing incident triage time by 40%.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built real estate data pipelines and property management modules (Python, Java, PostgreSQL); delivered large-scale data reconciliation workflows (1.6M HUD/Tax Credit records).
- Led AWS S3 cloud migration from legacy server storage; created 56 state-specific housing certification documents.

PROJECTS

AgentProbe -- LLM Backend with Vector Search Live on GitHub

Built a 15-endpoint FastAPI backend routing requests across five LLM providers with a **ChromaDB vector search layer** for semantic retrieval and a smart LLM Router for dynamic provider selection. CI regression tests with golden datasets (1,240+ test cases) run on every commit via GitHub Actions.

Python · FastAPI · ChromaDB · LangGraph · LangChain · Anthropic · OpenAI · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

Leasel -- Hybrid Search Pipeline over Real Estate Documents

Built a **hybrid search pipeline** (hand-rolled TF-IDF + semantic reranking) over lease and compliance documents -- section-aware chunking, OCR ingestion, cosine similarity ranking -- with answers cited to the exact source clause. 8-agent LangGraph workflow for support ticket investigation with HITL escalation. Tested with LLM-as-judge: 93.6% precision, 87.9% F1 across 1,240 test cases.

Python · LangGraph · FastAPI · TF-IDF RAG · Hybrid Search · OCR · SQLite · CI/CD -- github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR -- Clinical Data Search & REST API Integration

Built a full-stack search API translating natural-language questions into FHIR REST API queries via LLMs and BERT embeddings -- cut manual clinical lookup time by ~60%. Flask backend, React/TypeScript frontend, Docker, HIPAA-style RBAC.

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs -- github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University May 2026

GPA 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- JNTU Anantapur 2017

TECHNICAL SKILLS

Backend & APIs: Python, Node.js, FastAPI, Flask, Django, Java, Spring Boot, TypeScript, JavaScript, React, GraphQL, REST APIs, Microservices, SOAP

Search & Data: Elasticsearch, ChromaDB, Pinecone, Weaviate, TF-IDF, Semantic Search, Hybrid Search, Embeddings, PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization

AI & LLMs: Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, RAG Pipelines, LangGraph, LangChain, BERT, spaCy, HuggingFace, scikit-learn, XGBoost, PyTorch

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions, Jenkins CI/CD

Practices: Agile/Scrum, TDD, Code Review, REST API Design, Observability, HIPAA/PHI Compliance, Async Processing (RabbitMQ, Kafka, Celery)